NanoGap CMUT

Sub-100 nm-Gap Capacitive Ultrasonic Transducer Arrays
as Building Blocks for Ultrasound Neurotechnology

Baxter Barlow · Alan Duan · Rohan Dhillon · Eric Yee · October 27, 2026

Idea and Motivation

A capacitive micromachined ultrasonic transducer (CMUT) is a thin membrane suspended over a small air gap above an electrode (Fig. 1). A voltage pulls the membrane toward the electrode, allowing the device to emit ultrasound (electrically driven) and detect it (via changes in capacitance). CMUTs are promising for neurotechnology, where ultrasound is used both to image brain activity [1] and stimulate the cortex [2], because they can be lithographically patterned and integrated with electronics.

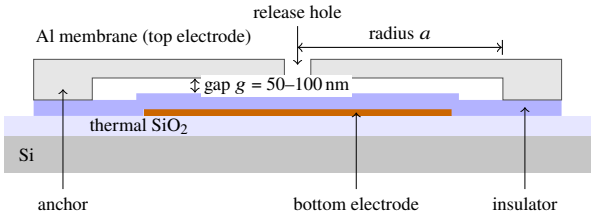


Figure 1: One CMUT cell in cross-section (not to scale). The gap is set by a thin photoresist layer that is dissolved through the release hole. A voltage between the membrane and the bottom electrode pulls the membrane toward the substrate.

The key design choice in our proposal is the gap: a smaller gap lowers the voltage required to drive the membrane. Still, it increases its susceptibility to sticking to the substrate during fabrication [3, 4]. A commercial ultrasound-on-a-chip uses a ~200 nm gap [5]. We will build CMUTs with 50, 75, and 100 nm gaps and measure how this trade-off plays out below 100 nm. Based on plate models and preliminary reports, we expect that reducing the gap will significantly decrease the pull-in voltage, making low-voltage operation feasible. However, we also anticipate a sharp increase in stiction as the gap narrows, particularly below 75 nm, which may limit the maximum usable membrane size. We hypothesize that optimal low-voltage performance will be balanced against a higher risk of stiction at the smallest gaps, revealing new regimes not accessible in previous work.

Objectives

- **O1 Fabricate** CMUT cells and small (8-element) arrays with 50, 75, and 100 nm gaps.
- **O2 Pull-in voltage.** Measure how the voltage at which the membrane snaps down depends on gap and membrane size, and compare with a simple plate model, $V_{PI} \propto g^{3/2}/a^2$.
- **O3 Stiction.** Find the largest membrane that survives release at each gap (predicted to grow as $\sqrt{g}$).
- **O4 Arrays.** Measure array uniformity and estimate its effect on steering an ultrasound beam.

Approach

Every die has the same test structures, so each fabrication run is also a measurement. Table 1 lists the steps each one uses.

- **Membranes of several radii.** The largest that survives release measures the stiction limit.
- **Insulator-free cells.** Touch-down shorts the cell, so pull-in and stiction show up electrically.
- **Test chips** processed alongside calibrate the thin resist films and test the insulator.
- **Control**: a ~1 µm-gap version built with the same process.

Characterization

Table 2 provides an overview of each measurement, its tool, and objective. We intend to perform all tests in air and characterize the devices electrically, without rigorously measuring acoustic output unless time permits.

Table 1: Fabrication sequence.

	Step	Process
1	Bottom electrodes	Pattern, evaporate Cr/Au, lift off
2	Insulator	Pattern, deposit 50 nm SiO_2, lift off
3	Gap layer	Spin 50/75/100 nm diluted resist; open anchor rings
4	Membrane	Evaporate 1 µm Al; pattern release holes and pads
5	Release	Dissolve resist in acetone; rinse in IPA; dry
6	Test	3D-print a chip holder; test; second run if time permits

Table 2: What we measure and how.

Measurement	Tool	Obj.
Gap and film thickness	Thin-film reflectance; SEM of cross-sections	O1
Release progress	Optical microscope	O1, O3
Pull-in voltage; stuck membranes	I–V setup	O2, O3
Array capacitance and element yield	I–V setup (capacitance–voltage sweeps)	O4
Insulator breakdown (test capacitors, first)	I–V setup	—

Materials and Tools

Materials

- Oxidized silicon pieces
- Cr, Au, Al, and SiO_2 deposition sources
- AZ3312 photoresist, thinner, developer
- Al etchant, acetone, IPA, 3D-printing resin

Tools

- MLA, evaporator, sputterer
- Spin coater, wet bench, 3D printer
- Optical microscope, SEM
- Optical characterization setup, I–V setup

References

[1] W. S. Griggs *et al.*, *Nat. Neurosci.* **27**, 196 (2024).
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[3] K. Brenner *et al.*, *Micromachines* **10**, 152 (2019).
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[5] J. M. Rothberg *et al.*, *Proc. Natl. Acad. Sci.* **118**, e2019339118 (2021).